

For rest of today: $E \subset \mathbb{R}^n$ is open, $f: E \rightarrow \mathbb{R}^n$ is loc. Lip., $I_x \subset \mathbb{R}$ is the maximal interval for any $x \in E$ (w.r.t. $\dot{y} = f(y)$, $y(0) = x$),

$$\Omega = \bigcup_{x \in E} I_x \times \{x\} = \{(t, x) \mid x \in E, t \in I_x\} \subset \mathbb{R} \times E,$$

and $\phi: \Omega \rightarrow E$ is the flow of f .

Def We say that f is **forward complete** if $I_x \supset [0, \infty)$ $\forall x \in E$ (or equivalently $\Omega \supset [0, \infty) \times E$), **backward complete** if $I_x \supset (-\infty, 0]$ $\forall x \in E$ (or equivalently $\Omega \supset (-\infty, 0] \times E$), and **complete** if f is both forward & backward complete (i.e., $I_x = \mathbb{R} \forall x \in E$, or equiv. $\Omega = \mathbb{R} \times E$).

Prop f is complete $\iff \phi$ is a global flow (i.e. $\Omega = \mathbb{R} \times E$).

Lemma

Let $x \in E$, $a, b \in (0, \infty)$ be s.t. $[-a, b] \subset I_x$.

("continuous dependence of
cpet subintervals
of max intervals"
+ "uniform compact
containers")

Then $\exists \delta > 0$ and a compact set $K \subset E$ s.t.,

$$\forall y \in \bar{B}_\delta(x), \quad [-a, b] \subset I_y \quad \text{and} \quad \phi^{[-a, b]}(y) \subset K.$$

pf

Given $r > 0$, define

$$A_r := \{p \in \mathbb{R}^n : \|p - \phi^t(x)\| \leq r \text{ for some } t \in [-a, b]\}.$$

Compactness of $[-a, b]$ and continuity of $t \mapsto \phi^t(x)$ imply that

$\phi^{[-a, b]}(x) (= \phi([-a, b] \times \{x\})) \subset E$ is compact, so $\exists \varepsilon > 0$ s.t.

A_ε is a compact subset of E . Thus, f is L -Lipschitz on

A_ε for some $L > 0$.

Choose $\delta > 0$ s.t. $\delta < \varepsilon e^{-L(a+b)}$ (hence $\delta e^{L(a+b)} < \varepsilon$).

Consider any $y \in \bar{B}_\delta(x) \subset A_\varepsilon$. Let $c, d > 0$ be s.t.

$$I_y = (-c, d).$$

Define $\bar{d} := \sup \{t \in [0, d) : \phi^{[0, t]}(y) \subset A_\varepsilon\} \leq d$.

The final lemma of lecture 17 implies that

$$\forall t \in [0, \bar{d}]: \|\phi^t(x) - \phi^t(y)\| \leq \|x - y\| e^{Lt} \leq \delta e^{Lt}. \quad (*)$$

If $\bar{d} \leq b$, then $\delta e^{L\bar{d}} \leq \delta e^{Lb} < \delta e^{L(a+b)} < \varepsilon$, by our choice of δ so (*)

$\Rightarrow \phi^{[0, \bar{d})}(y) \subset A_{\delta e^{Lb}} \subset \text{int}(A_\varepsilon)$. Compactness of $A_{\delta e^{Lb}}$ & the corollary

from lecture 17 $\Rightarrow \bar{d} < d$ (hence $\bar{d} \in I_y$), so continuity of $t \mapsto \phi^t(y)$ further

implies that $\phi^{[0, \bar{d}]}(y) \subset A_{\delta e^{Lb}} \subset \text{int}(A_\varepsilon)$. Since I_y and $\text{int}(A_\varepsilon)$ are open

& $t \mapsto \phi^t(y)$ is cont., $\exists K > 0$ s.t. $\bar{d} + K \in I_y$ & $\phi^{[\bar{d}, \bar{d}+K]}(y) \subset \text{int}(A_\varepsilon) \subset A_\varepsilon$,

contradicting the defn of $\bar{d}$. Thus, $\bar{d} > b$ and hence also $d > b$, with the former implying that $\phi^{[0,b]}(y) \subset A_\varepsilon$.

A similar argument shows that $c > a$ and $\phi^{[-a,0]}(y) \subset A_\varepsilon$.

Thus, $[-a,b] \subset I_y$ and $\phi^{[-a,b]}(y) \subset A_\varepsilon$. Taking $K := A_\varepsilon$ completes the proof. $\square$

Theorem Ω is open and $\phi: \Omega \rightarrow E$ is locally Lipschitz (hence continuous).

pf Consider any $(t, x) \in \Omega$. Then $t \in I_x$.

Since I_x is open and $0 \in I_x$, $\exists a, b > 0$ s.t.

$t \in (-a, b) \subset [-a, b] \subset I_x$. The lemma above $\Rightarrow \exists \delta > 0$ and

a compact set $K \subset E$ s.t. $[-a, b] \times \overline{B}_\delta(x) \subset \Omega$ and $\phi([-a, b] \times \overline{B}_\delta(x)) \subset K$. Since f is locally Lipschitz, $\exists L, M > 0$ s.t. f is L -Lipschitz on K and $\|f(p)\| \leq M \quad \forall p \in K$.

Thus, $\forall y, z \in B_\delta(x)$ and $s, r \in (-a, b)$,

$$\|\phi^s(y) - \phi^r(z)\| \leq \|\phi^s(y) - \phi^s(z)\| + \|\phi^s(z) - \phi^r(z)\|$$

by final lemma of lecture 17 $\longrightarrow \leq \|y - z\| e^{L|s|} + \left\| \int_r^s f(\phi^\tau(z)) d\tau \right\|$

$$\leq \|y - z\| e^{L|s|} + M \cdot |s - r|$$

$$\leq (e^{L(a+b)} + M) (\|y - z\| + |s - r|)$$

By Cauchy-Schwarz $\longrightarrow \leq \sqrt{2} (e^{L(a+b)} + M) \|(y - z, s - r)\|$.

or just invoke "norm equivalence" which yields a const $C > 0$ instead of $\sqrt{2}$

Thus, ϕ is Lipschitz on $(-a, b) \times B_\delta(x)$, an open set containing (t, x) (w/ Lipschitz const $\sqrt{2} (e^{L(a+b)} + M)$). Since $(t, x) \in \Omega$ was arbitrary, ϕ is locally Lipschitz. $\square$

Notation

Given $t \in \mathbb{R}$, define

$$\Omega_t := \{x \in E \mid (t, x) \in \Omega\}.$$

Lem

For each $t \in \mathbb{R}$, Ω_t is an open (possibly empty) subset of E .

pf

Consider arbitrary $t \in \mathbb{R}$. If Ω_t is empty, we are done. If Ω_t is not empty, then consider any $x \in \Omega_t$, so $(t, x) \in \Omega$. Since Ω is open (by the above Prop), $\exists \gamma, \delta > 0$ s.t. $(t - \gamma, t + \gamma) \times \bar{B}_\delta(x) \subset \Omega$. Thus, $B_\delta(x) \subset \Omega_t$, so Ω_t is open since $x \in \Omega_t$ was arbitrary. $\square$

Prop

Consider $t \in \mathbb{R}$ s.t. Ω_t is nonempty. Then
 $\phi^t(\Omega_t) = \Omega_{-t}$ and $\phi^t: \Omega_t \rightarrow \Omega_{-t}$ is
a homeomorphism (i.e., a continuous bijection w/ a continuous
inverse), and the inverse is $\phi^{-t}: \Omega_{-t} \rightarrow \Omega_t$.

Pf

Next time...